

CFR Intern	American Express	Jan 2021 - Current
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- Ongoing internship in the Credit and Fraud Business Division. Responsible for automating business critical tasks using Python, selenium, beautifulsoup and cron, on both Unix and Windows based systems.

iOS Developer, Intern	CS Soft Solutions Pvt. Ltd.	Summer 2019
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- Co-ordinated with the core team to develop a custom implementation of High Dynamic Range (HDR)
- Developed and deployed a Food Delivery app based on the user stories
- Developed an app utilizing Apple ARKit, for a furniture store where people could place virtual 3D models of various items in their room to see how it would look and make better decisions.

Chandigarh, India	University Institute of Engineering and	Jul 2017 – Jun 2021
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- B.E. in Information Technology. GPA: 8.2/10
- Coursework: Data Structures & Algorithms, Database Design and Management, Calculus IV, Linear Algebra, Statistics, Operating Systems, Computer Architecture, Theory of Computation & Compiler Design, Advanced Computer Networking, C/C++, Python

Remote **Udacity** **Nov 2018 – May 2019**

- Deep Learning Nanodegree – Credential ID: C9HLTFDN using PyTorch, Tensorflow, Keras
- Full Scholarship by Facebook, chosen out of 10,000 global applicants of diverse qualifications with an acceptance rate of 3%
- Coursework: Computer Vision, Sentiment Analysis, Generative Adversarial Networks, Deployment

SAT Score: 1390/1600 in May 2016 – 800/800 in Maths, 590/800 in English, 97th Percentile

PROJECTS

- **[Hadoop/Spark] Distributed Computing for Cyber Security:** Practically verified a theoretical technique based on Shannon / General entropy metrics to distribute the network analysis load over a computing cluster to identify and deal with DDOS attacks over WAN using Apache Hadoop & Spark.
- **[Brain Computer Interfacing] Cognitive Neuroscience:** Devised a method to detect muscle artefacts in EEG data using Brain-Computer Interfacing, with a potential for upgraded Brain Mapping by attenuation of the recognized artefacts. It was tested on the Left Frontal Lobe with >95% accuracy
- **[Computer Vision] Accident Detection:** Uses camera feed at traffic intersections to spot road accidents in real time.
- **[Software Development] iOS App for Resellers:** Scans product barcodes, aggregates the price history from different sources, so the reseller can make an informed decision. Deployed to the Apple App Store
- **Network Security:** Wrote Snort IDS rules to identify and handle intrusions in Panjab University's Network
- **Dark Matter Discovery:** Built a pipeline to visualize and confirm presence of Dark Matter in Cosmic Data based on the principle of Strong Gravitational Lensing, using PyAutoLens
- **Cosmic Artefact Filter:** Trained a deep learning model to flag cosmic ray artefacts in satellite imagery for deeper analysis. Used image data from NASA's MESSENGER Spacecraft orbiting the planet Mercury.
- **Quantum Machine Learning:** Researched and demonstrated use of Quantum Machine Learning to distinguish between signal and background events in High Energy Physics.

VOLUNTEER & ADDITIONAL EXPERIENCE

- **Speaker & Mentor:** Moderated Programming Club, UIET. Taught and guided junior CS students
- **Udacity Mentor:** Facilitated smooth flow of Deep Learning Nanodegree program on Udacity's behalf

Skills & Technologies

- C/C++, Python, Java, JavaScript, Version Control and Code Collaboration with Git
- ReactJS, Xcode, NodeJS, MySQL, Firestore, PyTorch, Tensorflow, AWS
- Full Stack Web Development, iOS App Development, Machine Learning, RESTful APIs, UNIX Systems